

Benchmark design factors

How the **design and evaluation variables** relate in the sceptical-llms base-rate benchmarks.

Viewing diagrams: Cursor/VS Code built-in preview (**Ctrl+Shift+V**) does **not** render Mermaid. Use the **SVG figures** below (they show in preview and print), open this file on **GitHub**, or install the [Markdown Preview Mermaid Support](#) extension. Diagram sources: [docs/figures/benchmark-design-factors-*.mmd](#).

Two layers:

Layer	Where	What it indexes
Prompt design	data/*/benchmark.csv	One row per example_id (vignette × variant × disclosure fork)
Evaluation results	merged CSVs, Kaggle *.run.json	One row per (example_id , model) — every design cell crossed with each evaluated LLM

Main variables

Six fields stratify almost all analysis.[^derived]

Variable	Layer	Role
vignette_name	design	Which scenario (content domain, probabilities, wording)
variant	design	Response format and option menu
problem_type	design	Partition vs overlap (and disclosure) vs implausible
scepticism_required	design	Whether the keyed correct answer is sceptical/meta rather than blind Bayes
intersection_size	design	Strength of C∩D overlap (0 / small / medium / large)
model	evaluation	Which LLM produced the response (Kaggle model slug)

Observational unit (merged results): ([vignette_name](#), [variant](#), [problem_type](#), [model](#)) — equivalently ([example_id](#), [model](#)), since [example_id](#) encodes vignette × variant × disclosure fork.

[^derived]: [response_type](#) is derived from [variant](#) (base rate: [open_probs](#) → [open](#), [mc_numeric_probs](#) → [mc_numeric](#), [mc_full_probs](#) → [mc_full](#); simple CSV stores [mc](#) for MC rows, normalized to [mc_numeric](#) at load). [has_statistics](#) is currently always [true](#) (no [*_no_probs](#) variants are built). Prefer [variant](#) over [response_type](#) in groupbys.

Factor definitions

[vignette_name](#) (content — comes from the vignette source)

Human-readable scenario label (e.g. `CA Trump voter`, `college STEM work`). Fixed per story; crossed with `variant` (and, in base rate, overlap-disclosure / implausible forks) to form `example_id`.

- `intersection_size` is fixed per vignette (see below).
- `problem_type` / `scepticism_required` are determined by vignette class and disclosure pass (base rate only varies these across forks of the same name).
- Nine vignettes in each of simple and base-rate builds today; simple reuses overlap vignettes with explicit overlap in the narrative.

Full list and overlap sizes: `docs/vignette-table.txt`.

`intersection_size` (structural — comes from the vignette source)

Measures overlap between pathways C and D under universe A:

Ratio = $P(C \cap D | A) / \max(P(C | A), P(D | A))$

Label	Meaning
<code>∅</code>	Partition vignettes (two-cause / well-posed); $P(C \cap D)$
<code>small</code>	Overlap ratio < 10%
<code>medium</code>	10%–25%
<code>large</code>	> 25%

Set in source CSVs (`docs/base-rate-two-cause-vignettes.csv`, `docs/base-rate-overlap-vignettes.csv`) and propagated into items. **Does not vary within a vignette** — it is a property of the story, not of `variant`.

`problem_type` (derived from vignette + overlap disclosure)

Value	When
<code>well_posed</code>	Partition vignette (<code>well_posed=True</code> on source row)
<code>overlap_explicit</code>	Overlap vignette; prompt states $P(C \cap D)$
<code>overlap_implicit</code>	Overlap vignette; overlap not stated — solver must infer or sceptically decline
<code>implausible</code>	Partition vignette with statistics replaced from <code>implausible_statistics.csv</code>

Logic (`scripts/build_base_rate_prompts.py` → `_problem_type`):

- Implausible fork → `implausible`
- Else if partition → `well_posed`
- Else → `overlap_{explicit|implicit}` from the disclosure pass

Simple benchmark: always `well_posed` (partition + overlap vignettes are both included, but overlap is always explicit in the simple two-path narrative; no implicit/implausible forks).

`scepticism_required` (derived — scoring expectation)

Value	When
true	Implausible vignette or overlap vignette with implicit disclosure
false	Partition vignette or overlap with explicit disclosure

Logic (`scepticism_required()` in `build_base_rate_prompts.py`):

implausible	→ true
overlap + implicit disclosure	→ true
overlap + explicit disclosure	→ false
well_posed (partition)	→ false

Simple benchmark: always `false`.

When `scepticism_required=true`, the normative response is a **meta** answer (insufficient / inconsistent / obviously incorrect), not the overlap-aware Bayes posterior.

`variant` (response format — crossed with every applicable vignette × disclosure cell)

Variant	Prompt shape	MC options
<code>open_probs</code>	Free-text percentage	—
<code>mc_numeric_probs</code>	Multiple choice, numeric lures only	A–E (count varies: 4 or 5)
<code>mc_full_probs</code>	Multiple choice, numeric + meta	A–E plus F, G, H (insufficient / inconsistent / obviously incorrect)

Which variants are emitted (`variants_for_vignette`):

<code>scepticism_required = true</code>	→ <code>mc_full_probs</code> only
<code>scepticism_required = false</code>	→ <code>open_probs</code> , <code>mc_numeric_probs</code> , <code>mc_full_probs</code>

Simple benchmark: only `open_probs` and `mc_numeric_probs` (no F/G/H). See [[^]derived] for `response_type`.

`model` (evaluation — crosses every design cell)

Not in `benchmark.csv`; added when prompts are run on Kaggle (or locally) and merged into results.

Column	Source
<code>model</code>	Kaggle <code>modelVersion.slug</code> from each <code>*.run.json</code> (e.g. <code>google/gemini-3-flash-preview</code> , <code>openai/gpt-5.5-2026-04-23</code>)

Properties:

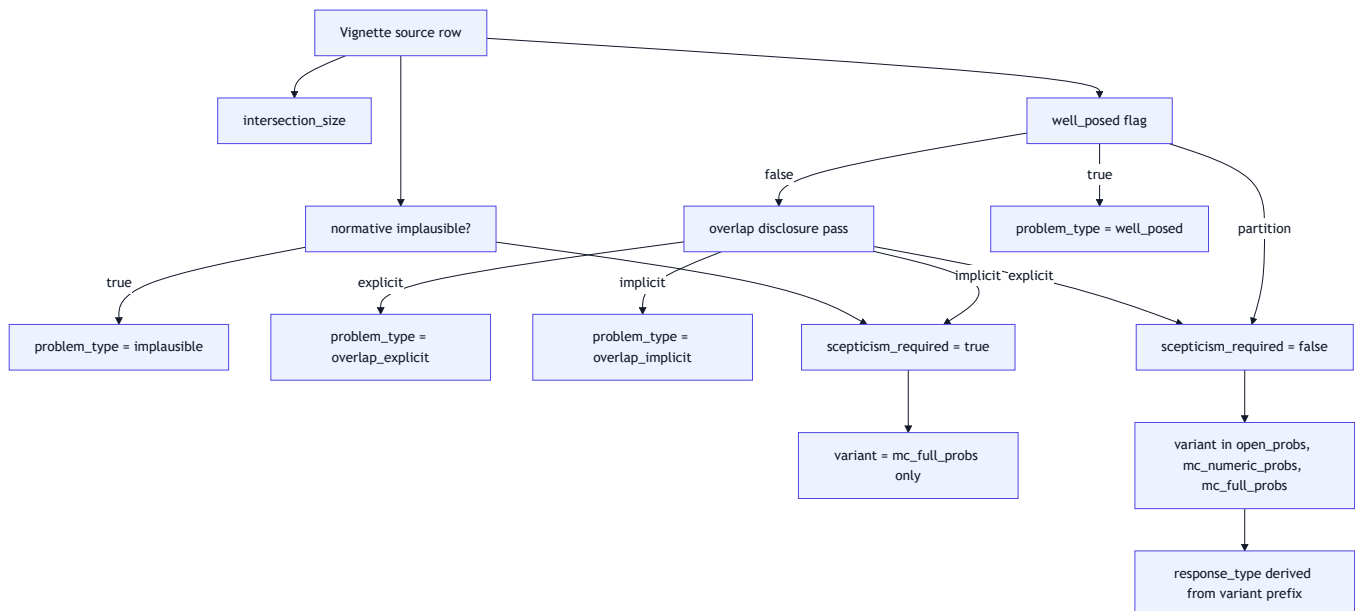
- **Independent of design factors** — every `example_id` is evaluated once per model in the run grid; `model` does not change the prompt or scoring keys.
- **Full factorial with design** — merged rows are the Cartesian product of design items × models in the task run (e.g. simple: 18 prompts × 6 models → 108 rows).
- **Dedup key** — when loading multiple downloads, `dedupe_base_rate_prompt_records` keeps the **last** record per (`model`, `example_id`) (`benchmarks/kaggle_runs.py`).
- **Not in notebook cache id** — `kbench` caches by (`evaluation row index`, `model`) only; changing prompts without clearing `*.run.json` can reuse stale answers for the same model + row index.

Current simple-benchmark run (example): 6 models × 18 `example_ids` = 108 merged rows.

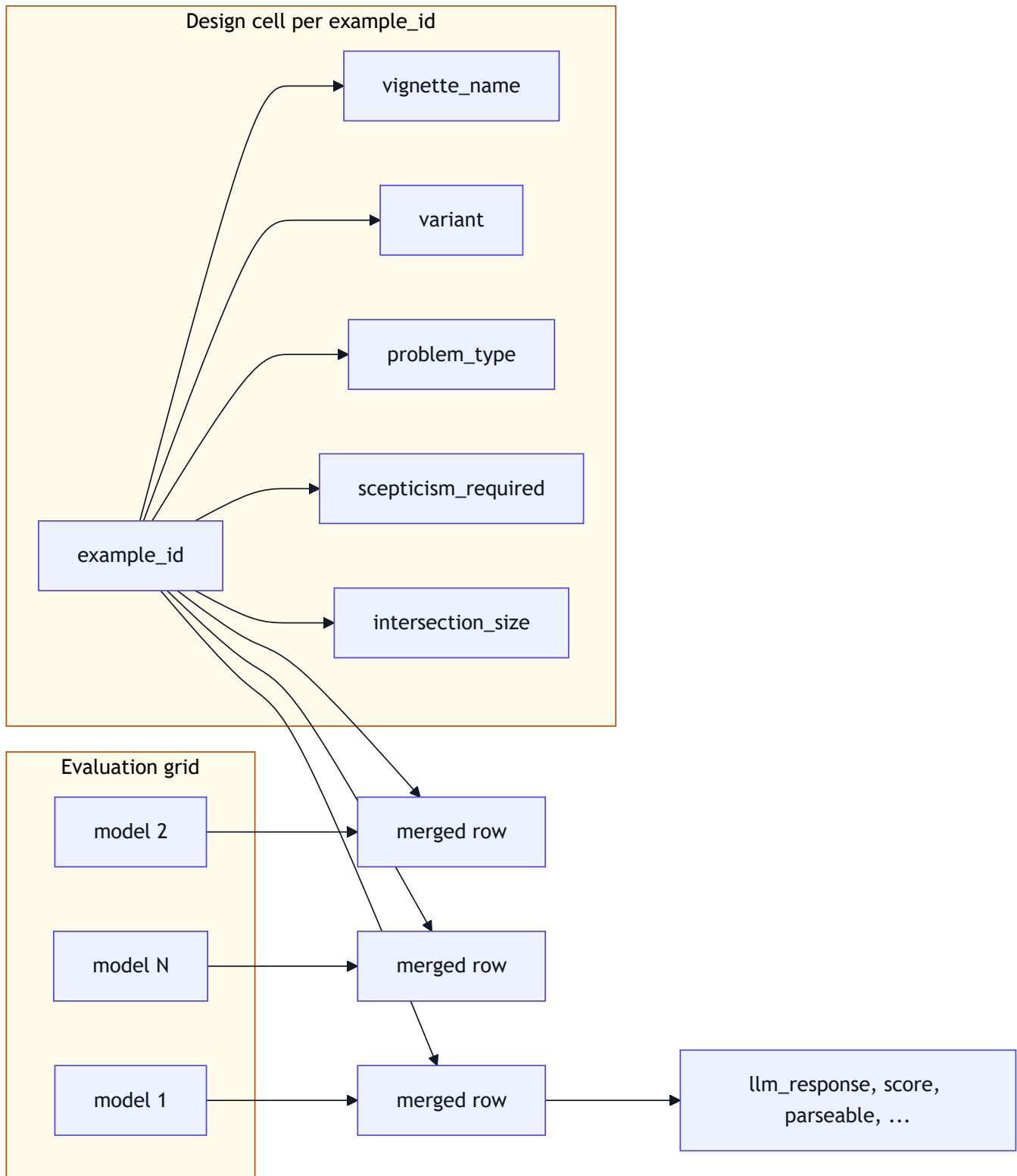
Dependency diagram

Factors are **not independent**. Arrows read as “determines” or “constrains”.

Design factors (prompt CSV)



Evaluation layer (design × model)



Each `example_id` is replicated once per evaluated `model`; outcomes (`llm_response`, `score`, `path_c_confusion`, etc.) vary by model, not by design rules.

Re-render after editing `.mmd` sources (white background + high-contrast arrows for dark IDE preview):

```

cd docs/figures
npx @mermaid-js/mermaid-cli -c benchmark-design-mermaid-config.json \
  -i benchmark-design-factors-design.mmd -o benchmark-design-factors-design.svg -b
white
npx @mermaid-js/mermaid-cli -c benchmark-design-mermaid-config.json \

```

```
-i benchmark-design-factors-eval.mmd -o benchmark-design-factors-eval.svg -b white
```

Derived / redundant columns

Column	Relationship
response_type	Derived from variant — see [^derived]
has_statistics	Currently always true — see [^derived]
example_id	Slug encoding vignette_name + variant + disclosure/implausible fork
well_posed / normative (scoring)	Align with problem_type but used for scoring keys, not identical labels
scepticism_score_target	Scoring key that depends on variant + scepticism_required (percent, letter, F\ G\ H , meta , etc.)

Actual combinations in the repo (current CSVs)

Simple benchmark (**data/simple/benchmark.csv**) — 18 rows

9 vignettes × 2 variants. Single cell in factor space:

problem_type	scepticism_required	intersection_size	variants
well_posed	false	0 (5 vignettes), small (1), medium (1), large (2)	open_probs , mc_numeric_probs

No **overlap_***, **implausible**, or **mc_full_probs**.

Base-rate benchmark (**data/base_rate/benchmark.csv**) — 44 rows

problem_type	scepticism_required	intersection_size	variants
well_posed	false	0	open, mc_numeric, mc_full (5 vignettes each → 15 rows)
overlap_explicit	false	small, medium, large	open, mc_numeric, mc_full (4 overlap vignettes → 12 rows)
overlap_explicit	true	small, medium, large	mc_full only (4 rows)
overlap_implicit	true	small, medium, large	mc_full only (8 rows)
implausible	true	0	mc_full only (5 rows)

Cross-tab: **scepticism_required** × **variant** (base rate)

	open_probs	mc_numeric_probs	mc_full_probs
false	9	9	9
true	0	0	17

When scepticism is required, only the full MC menu (with F/G/H) is offered.

Merged results (design × model)

Benchmark	Design rows	Models (example run)	Merged rows
Simple	18	6	108
Base rate	44	varies by download	44 × n_models

Row count formula: `n_example_ids × n_models` (plus empty response rows if a model did not complete an item).

What the main variables do *not* encode

- Exact probabilities and prompt text** — fixed by `vignette_name` but not repeated in the six factor names (see `benchmark.csv` / `items.csv`).
- Simple vs base-rate benchmark** — simple is a deliberate subset (two-path P(C|T), no scepticism fork, no `mc_full`).
- Overlap disclosure** as its own column — folded into `problem_type` (`overlap_explicit` vs `overlap_implicit`).
- Implausible parameter edits** — only distinguished by `problem_type=implausible + intersection_size=0`; which statistic was perturbed is in source CSVs.
- Lure layout** (which letters map to normative vs P(T|C) confusion, etc.) — scoring metadata (`option_*_lure`), not a top-level factor.
- Run metadata** — `reasoning`, token limits, task version, run timestamp (in Kaggle JSON, not always in merged CSV).

Suggested analysis groupings

Question	Group by
Which scenario	<code>vignette_name</code>
Open vs MC numeric vs MC full	<code>variant</code>
Partition vs overlap vs implausible	<code>problem_type</code>
Should model sceptically refuse?	<code>scepticism_required</code>
Overlap strength	<code>intersection_size</code> (within overlap / well_posed rows)

Question	Group by
Model comparison	<code>model</code> (rows or columns in pivot tables)
Model × condition interaction	<code>model</code> × <code>variant</code> (or × <code>problem_type</code> , etc.)
Full design cell	<code>(vignette_name, variant, problem_type)</code>
Single observational unit	<code>(example_id, model)</code> or <code>(vignette_name, variant, problem_type, model)</code>
Content-only comparison	<code>vignette_name</code> holding <code>variant</code> and <code>model</code> fixed

Code references

Logic	File
Factor definitions, variant gating, scepticism rules	<code>scripts/build_base_rate_prompts.py</code>
Simple subset (2 variants, well_posed only)	<code>scripts/build_simple_rate_prompts.py</code>
Intersection size thresholds / vignette list	<code>docs/vignette-table.txt</code>
Model slug from Kaggle runs, merge/dedupe	<code>benchmarks/kaggle_runs.py</code>
Per-model evaluate grid	<code>benchmarks/simple_rate_tasks.py</code> , <code>benchmarks/base_rate_tasks.py</code>